

7.4.3

AI25BTECH11036-SNEHAMRUDULA

Question:

Q. From the origin chords are drawn to the circle

$$\left(u - \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right)^T \left(u - \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = 1 \quad (0.1)$$

The equation of the locus of the mid-points of these chords is to be found.

Solution:

1.The given circle equation is

$$\mathbf{u}^T \mathbf{u} - 2 \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{u} = 0 \quad (0.2)$$

which can be expressed in standard form as

$$\mathbf{u}^T \mathbf{u} + 2\mathbf{c}^T \mathbf{u} + f = 0 \quad (0.3)$$

where

$$\mathbf{c} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad f = 0 \quad (0.4)$$

2.A chord of this circle passing through the origin $O = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ has midpoint $\mathbf{m}$. From the property of chord bisectors:

$$\mathbf{m}^T \mathbf{c} + \mathbf{m}^T \mathbf{m} = 0 \quad (0.5)$$

3.Substituting $\mathbf{c} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$,

$$-m_1 + m_1^2 + m_2^2 = 0 \quad (0.6)$$

4.Hence, the locus of $\mathbf{m}$ is

$$\mathbf{m}^T \mathbf{m} = m_1 \quad (0.7)$$

Thus, the locus of the midpoints is the circle

$$m_1^2 + m_2^2 = m_1,$$

i.e., a circle with center $\left(\frac{1}{2}, 0\right)$ and radius $\frac{1}{2}$.

